BE (E&TC)	PLC SCADA and Automation (Elective-III)	
Experiment No: 4	To implement a traffic light control system using a PLC with timer instructions (Software/ Hardware Implementation).	**Pages**

AIM: To implement a traffic light control system using a PLC with timer instructions (Software/Hardware Implementation).

Software Used: TIA 13(Totally Integrated Automation)

OBJECTIVES:

1. Understand the working principle of PLC timers.
2. Learn to design and implement a traffic lightcontrol sequence.
3. Gain experience in PLC programming using ladder logic.

APPARATUS REQUIRED:

1. PLC (Programmable Logic Controller) – Siemens/Allen Bradley/Delta (or equivalent).
2. PLC Programming Software (e.g., TIA Portal, RSLogix, WPLSoft).
3. Traffic light model with Red, Yellow, and Green lamps (or LEDs).
4. Power supply (24V DC or as per PLC specification).
5. Connecting wires.
6. Computer/Laptop with PLC software installed.

THEORY:

Traffic light control is a common automation example to demonstrate PLC timer usage. A standard
sequence includes: - Green light ON for a set duration. - Yellow light ON for a short warning duration. - Red light ON for a set duration. Using PLC, timers such as TON (On-Delay Timer) are
used to control the ON and OFF times for each light in sequence.

PROCEDURE:

1. Connect the Red, Yellow, and Green lights to the PLC output terminals.
2. Assign input switches (if any) to start/stop the traffic light sequence.
3. Open PLC programming software and select the correct PLC model.
4. In Ladder Logic, create a sequence: - Start with Green light ON using a TON timer for 10 seconds. - Switch to Yellow light ON using a TON timer for 3 seconds. - Switch to Red light ON using a TON timer for 10 seconds.
5. Repeat the sequence in a loop.
6. Download the program to the PLC.
7. Run and observe the light sequence.

Ladder Logic Description:

- T1: Green light ON (Q0.0) for 15 seconds. - T2: Yellow light ON (Q0.1) for 5 seconds after T1.
T3: Red light ON (Q0.2) for 20 seconds after T2. - Cycle repeats continuously.

Expected Output:
Step Light ON Timer Duration
1 Green 15 seconds
2 Yellow 5 seconds
3 Red 20 seconds

Ladder diagram:

Symbolic Ladder Logic:

```
[ ] Start_PB   ---| |---[TON T1 5s]---( ) Red_Light

[ ] T1 Done    ---| |---[TON T2 2s]---( ) Yellow_Light

[ ] T2 Done    ---| |---[TON T3 5s]---( ) Green_Light

[ ] T3 Done    ---| |---(RES T1,T2,T3)
```

Explanation:

- Press Start_PB to initiate the traffic light sequence.
- T1 turns ON Red light for 5 seconds.
- After T1, T2 turns ON Yellow light for 2 seconds.
- After T2, T3 turns ON Green light for 5 seconds.
- After T3, all timers reset and cycle restarts.

CONCLUSION

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Questions:

1. Draw the basic block diagram of a PLC-based traffic light control system.

2. What is the working principle of a traffic light control system?

3. Why are timers used in traffic light control?

4. List the sequence of light changes in a standard traffic signal cycle.

REFERENCES:

[1]Petruzella, F. D. (2017). *Programmable Logic Controllers* (5th ed.). McGraw-Hill Education.
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[3] Siemens AG. (2023). *TIA Portal Programming Manual*.Siemens Industry Online Support